

What do you notice about the places?

Say, draw, point or write. Keep your first thinking so you can revisit it.

Work on your own first. For each comparison, write $<$, $>$ or $=$ and give a short reason. You may use a labelled place-value chart or explain aloud. Mark a question you want help with; this helps us choose what to learn next.

1. What is the value of the digit 4 in 58,406? What is its value in 3.406? Name each place.

2. 45,082 __ 45,208. Name the first place that decides the comparison.

3. 6.4 __ 6.04. Explain what the 4 is worth in each number.

4. 2.5 __ 2.50. Explain using tenths and hundredths.

5. 0.507 __ 0.57. Name the first unequal place.

6. A learner says, '0.308 is greater than 0.8 because 308 is greater than 8.' Do you agree? Give a place-value reason.

Represent and explain the units

Show what each digit counts and explain a comparison or equality using the same units.

Worked example

Compare 2.4 and 2.04. Use the same whole for both numbers.

1. Use these columns: ones, tenths, hundredths, thousandths. For 2.400 the entries are 2, 4, 0, 0. For 2.040 they are 2, 0, 4, 0.
2. Both have 2 ones. Four tenths equals forty hundredths, so 2.4 is 2 wholes and 40 hundredths. The other number is 2 wholes and 4 hundredths.
3. Compare the tenths first: 4 tenths is greater than 0 tenths. Write $2.4 > 2.04$.
4. The final zeros in 2.400 and 2.040 help align the places; they add no extra value. The zero between the decimal point and the 4 in 2.04 holds the tenths place.

So: $2.4 > 2.04$. The tenths are the first unequal place; 40 hundredths is greater than 4 hundredths.

Now try

1. Write 6.205 as a sum of its non-zero place values. What is the value of the 5?

2. $0.7 _ 0.70$. Express both numbers in hundredths and explain.

3. Write the ones, tenths, hundredths and thousandths in 0.406. Then compare 0.406 and 0.46.

Explain one answer to a partner. Then use your route's questions on the fresh-check page independently.

MATHS | ONE USEFUL ROUTE

Compare from the greatest place

Compare and order whole numbers and decimals, keeping equal values together and justifying the first difference.

Worked example

Compare 7.305 and 7.35. Explain the first unequal place.

1. Write 7.35 as 7.350 so equivalent places line up. The value stays the same.
2. Read from the greatest place: 7 ones matches 7 ones; 3 tenths matches 3 tenths.
3. The hundredths first differ: 0 hundredths is less than 5 hundredths. Stop there: $7.305 < 7.350$.
4. The later 5 thousandths cannot overturn the earlier hundredths difference. Name the place in your explanation, not just the digit.

So: $7.305 < 7.35$. Hundredths decide: $0 < 5$.

Now try

1. 8,046 __ 8,406. Name the first unequal place.

2. 5.09 __ 5.9. Name the first unequal place.

3. Order 0.609, 0.69, 0.6 and 0.600 from least to greatest. Put equal values together and explain one comparison.

Explain one answer to a partner. Then use your route's questions on the fresh-check page independently.

Use comparisons to make a decision

Use precise decimal comparisons to meet conditions and justify a large-number comparison.

Worked example

Paper strips measure 1.205 m, 1.25 m and 1.250 m. Choose every strip that is at least 1.24 m long and less than 1.3 m long.

1. Use metres throughout. 'At least 1.24' includes 1.240; 'less than 1.3' excludes 1.300.
2. Compare 1.205 with 1.240. The hundredths decide: $0 < 4$, so that strip is too short.
3. Write 1.25 as 1.250. Both remaining lengths are equal, greater than 1.240 and less than 1.300.
4. Select both strips marked 1.25 m and 1.250 m. Explain the two limits as well as the equality.

So: 1.25 m and 1.250 m both meet the conditions. 1.205 m does not.

Now try

1. Ribbon lengths are 0.708 m, 0.78 m, 0.780 m and 0.807 m. Choose every length that is at least 0.75 m and less than 0.80 m. Explain each exclusion.

2. Use each digit 2, 4 and 7 exactly once after the decimal point: 0.____. Find every possible number greater than 0.42 and less than 0.48. Explain how you know your list is complete.

3. Compare 2,406,000,000 and 2,460,000,000. Which is greater, and which place first decides?

Explain one answer to a partner. Then use your route's questions on the fresh-check page independently.

Try fresh numbers

Complete only your chosen route's two questions. Work independently with your usual supports. Explain your thinking.

Represent and explain the units

1. On your own: $3.07 \underline{\hspace{0.5cm}} 3.7$. Show the places or explain what each 7 counts.

2. $0.6 \underline{\hspace{0.5cm}} 0.60$. Explain why your symbol is correct.

Compare from the greatest place

1. On your own: $9,072 \underline{\hspace{0.5cm}} 9,027$. Name the first unequal place.

2. $0.804 \underline{\hspace{0.5cm}} 0.84$. Name the deciding place and explain why a longer decimal is not always greater.

Use comparisons to make a decision

1. On your own: lengths are 2.095 m, 2.105 m and 2.150 m. Choose every length that is at least 2.10 m and less than 2.15 m. Explain the boundary cases.

2. Use each digit 1, 3 and 8 exactly once after the decimal point: $0.\underline{\hspace{0.5cm}}$. Find all possibilities strictly between 0.31 and 0.39. Explain.

Which idea comes next?

Stop here: your teacher reviews this page with you before the bridge page.

Before the worked model is shown, try the four mini-check questions on your own with your usual access supports. Show or explain your thinking. Your teacher will decide with you whether to try the bridge or strengthen one earlier idea first.

1. Find 6×7 . Explain with equal groups, an array or a known fact.

2. Find 23×10 . Explain what the 2 and the 3 are worth in the result.

3. Solve 24×3 in your own way. Show how you kept track of tens and ones.

4. Estimate 31×9 using 30×10 . Which possible answer is reasonable: 279 or 2,790? Explain without needing the exact calculation.

Count every part

Use this page after your teacher reviews the mini-check and chooses the bridge with you.

Worked example

There are 14 packs with 23 stickers in each. Find 23×14 by splitting 14 into $10 + 4$.

- 1.** Estimate first: $20 \times 15 = 300$, a nearby friendly product. Keep 14 groups of 23 in mind.
- 2.** Split the groups: 10 groups give $23 \times 10 = 230$; 4 groups give $23 \times 4 = 92$, because $80 + 12 = 92$.
- 3.** Combine every part: $230 + 92 = 322$ stickers. These are partial products because each counts part of the full set of groups.
- 4.** Check another split: $20 \times 14 + 3 \times 14 = 280 + 42 = 322$. The exact answer also makes sense near the estimate of 300.

So: $23 \times 14 = 322$ stickers. Both decompositions count all 14 groups of 23.

Practise, then try a fresh example

1. With a partner: 12 packs have 32 counters each. Solve 32×12 using partial products. Label what each partial product counts.

2. Fresh check, on your own: 13 trays hold 18 counters each. Estimate, then solve 18×13 with a visible strategy. Explain why each part belongs.
